

SE 216 – SOFTWARE PROJECT MANAGEMENT

SOFTWARE MEASUREMENTS DOCUMENT

PROJECT NAME: NeuroMatter

GROUP NUMBER and MEMBERS: Duru Deniz Alptekin, Şevval Acar, Kaan Efe Koygun, Ege Haktan Aykut, Ali Utku Aydın, Eren Karlı, Deren Kocabaş, Efe Berkay Soyucak

Questions to identify measurements:

- Did the project progress according to the planned sprint schedule? (e.g. Sprint 1: 23.03 – 08.05.2026)
- How much effort did the team spend on development, AI integration, and testing each week?
- How many defects were found in the local AI module, sensor pipelines, and Flutter UI during testing?
- How effective were the test cases for caregiver alert logic, YOLOv8n detection, and SQLCipher encryption?
- How many requirement changes occurred across sprints (e.g. ADHD thresholds, notification logic updates)?
- What percentage of planned features were actually delivered by the end of each sprint?

Identified measurements:

- Planned task completion date / Actual task completion date (per sprint milestone)
- Weekly development hours / Weekly testing hours per team member
- Number of defects found / Number of critical defects (e.g. sensor data loss, AI false alerts)
- Total test cases / Passed test cases / Failed test cases
- Local AI detection accuracy (% correct fall / anomaly detections)
- Number of requirement changes per sprint iteration
- Feature completion rate (delivered features / planned features per sprint)

Measurement storage and collection:

What: Development and testing hours – hours logged per team member each week

When: End of each week

Format: Real number (e.g. 14.5 hrs)

How: Each team member logs hours in a shared Google Sheet or Notion tracker

SE 216 – SOFTWARE PROJECT MANAGEMENT

SOFTWARE MEASUREMENTS DOCUMENT

What: Number of defects and critical defects (sensor pipeline, AI model, Flutter UI)

When: During or after each testing session

Format: Integer count with severity tag (critical / medium / low)

How: Entered by the tester into GitHub Issues with severity labels

Measurement Type	Description	Example Measurements
Schedule	Compares planned vs. actual sprint task completion date	SQLCipher task: planned 08.05.2026, completed 10.05.2026 (+2 days)
Effort	Weekly development and testing hours per team member, % of planned sprint features delivered by sprint end	18h development, 6h testing, 2 of 3 features delivered = 67% completion
Defect tracking	Defects found per sprint, classified by severity	5 defects — 2 critical (false fall alert, missed anomaly)
AI performance	% of correct detections by local YOLOv8n / Qwen2 model	18/25 correct detections = 72% accuracy
Test coverage	% of test cases passed out of all executed per sprint	20/25 passed = 80% pass rate
Change	Number of requirement changes logged per sprint	added ADHD reminder, modified caregiver threshold